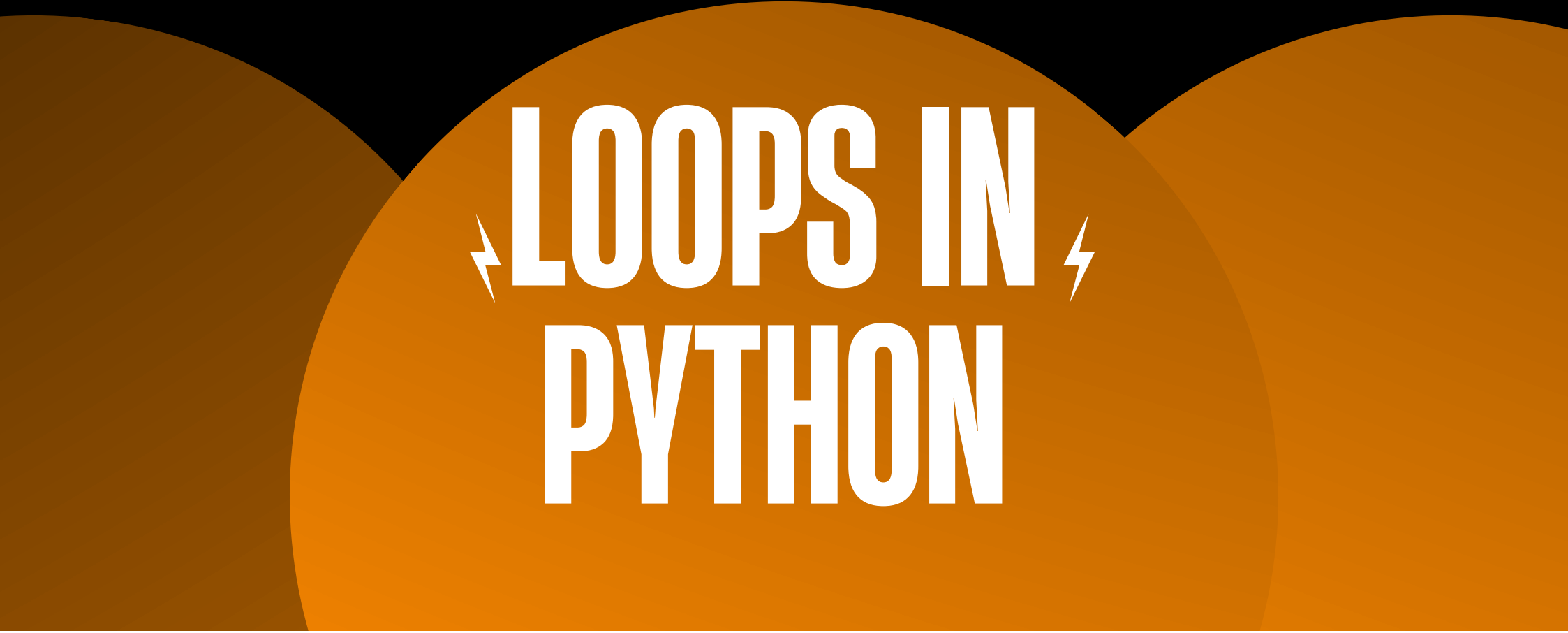


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⚡ LOOPS IN ⚡ PYTHON

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WHAT IS FOR LOOP

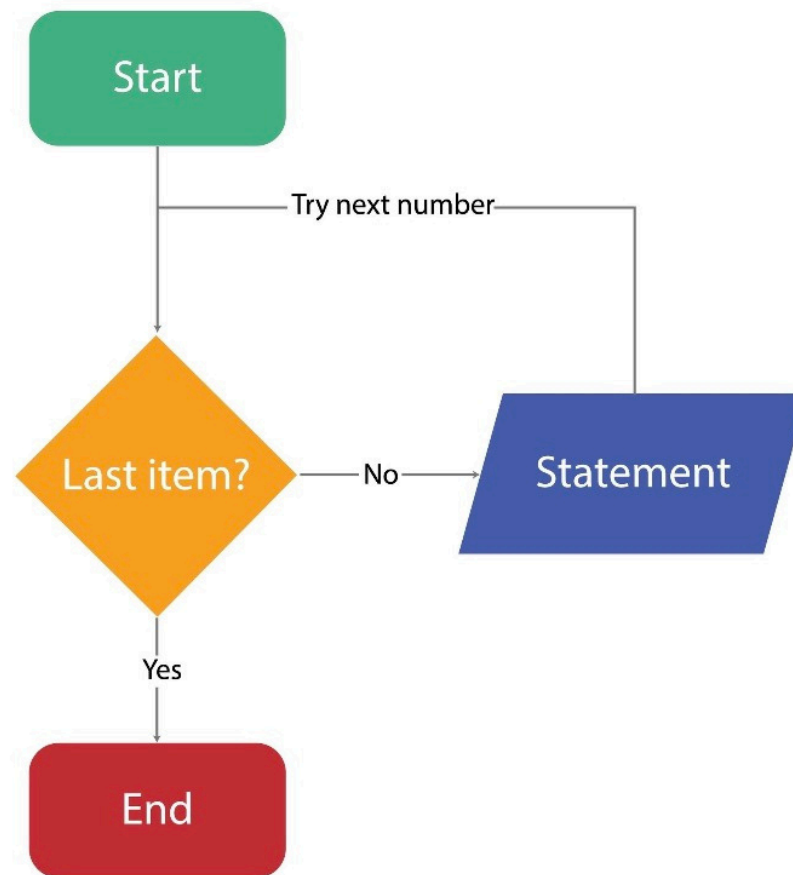
Definition: A control flow statement used to repeat a block of code a specific number of times.

- **The "Why":** Instead of writing the same line 100 times, you write it once and tell Python to repeat it.
- **Key Concept:** It "iterates" over a sequence (like a list, a string, or a range of numbers).

In Python, the syntax is designed to read like English:

- **for:** The keyword that starts the loop.
- **item:** A placeholder variable for the current element.
 - **sequence:** The collection you are looping through.
- **The Colon (:):** Essential to start the "body" of the loop.
- **Indentation:** All code inside the loop must be indented.

For Loop Flowchart



HOW IT WORKS EXAMPLE-

The Sequence: ["Alice", "Bob", "Charlie", "Diana"]

The Variable: name (this changes every time the loop runs).

The Output:

Done! Your coffee is ready, Alice.

Done! Your coffee is ready, Bob.

Done! Your coffee is ready, Charlie.

Done! Your coffee is ready, Diana.

All orders have been served

HOW IT WORKS EXAMPLE-

```
# Our list of friends
friends = ["Alice", "Bob", "Charlie", "Diana"]

# The for loop
for name in friends:
    print(f"    Done! Your coffee is ready, {name}.")

print("All orders have been served!")
```

THANKYOU

